

Supplementary Materials for: Predicting the long-term impact of AI assistants on collective intelligence

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Materials and Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors (Hong et al., 2012; Mann and Helbing, 2017; Wang et al., 2025):

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (\text{S1})$$

where the coefficient α_i determines the magnitude and direction of the effect of factor i , and α_0 represents a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i .

Players

We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a prediction task (Wang et al., 2025). In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A . Player A observes the causal factor X_{e_A} and makes a prediction by multiplying the observed factor value by their belief:

$$Y_A = c_A X_{e_A}, \quad (\text{S2})$$

where e_A can take an integer value from 0 to M . When $e_A = 0$, the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player's prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that $X_0 = 1$.

When the group is assisted by AI, each player's strategy additionally includes reliance on AI, denoted by β_A . We assume that each player consults the AI independently and revises their prediction by taking a weighted average of their own prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (\text{S3})$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i , and the stochastic error is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. For simplicity, we assume $b_i = b$ for the main analysis for all i . The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ . Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore hold τ constant and vary the magnitude of the additive bias to manipulate the level of AI accuracy during our analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (\text{S4})$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (\text{S5})$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous work (Wang et al., 2025), depending on the type of AI used by the group. First, when the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \cdots + Y_n^{\text{re}}}{N}. \quad (\text{S6})$$

Second, when the group is assisted by the Chatbot AI, both the player’s initial prediction and the AI’s prediction are based on the same causal factor. Each revised prediction therefore contains information about only one factor and can be treated as an estimate of that factor’s contribution to the target. We consequently use clustering aggregation.

Clustering aggregation is performed by grouping individuals according to their interests and calculating the mean prediction within each group. The collective prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (\text{S7})$$

where $\mathcal{A}_i$ represents the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ represents the number of players in that set.

When the group is not assisted by AI, collective predictions are computed from the unrevised individual predictions throughout the simulation. Both simple averaging and clustering can be used in the absence of AI. For fair comparison, we use the same aggregation rule as in the corresponding AI-assisted condition.

Incentive structures

Players receive payoffs based on the underlying incentive structure. We consider five incentive structures: feedback structure, niche-expert structure, direct intervention on AI use of these two structures, and a balanced structure.

The feedback structure does not directly reward individuals according to their own prediction accuracy. Instead, it rewards predictions that are aligned with the residual error of the collective prediction and therefore have the potential to move the collective prediction toward the true value. Given player A ’s revised prediction Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (\text{S8})$$

The niche-expert structure assigns higher payoffs to individuals who make accurate predictions while having an underrepresented interest. Specifically, the squared prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (\text{S9})$$

Previous research has shown that these two incentive structures can produce near perfect collective accuracy through social learning, except when the niche-expert incentive is combined with simple averaging (Wang et al., 2025). They are therefore used as baseline structures for examining how AI assistance affects the emergence of CI.

99 To examine the effects of directly encouraging or discouraging AI use, we mod-
 100 ify the feedback and niche-expert structures by adding a term proportional to the
 101 player’s reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \quad (\text{S10})$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (\text{S11})$$

102 Here, λ represents the strength of the direct incentive or penalty applied to AI use. A
 103 positive value of λ rewards AI reliance, whereas a negative value penalizes it. When
 104 $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to
 105 the original feedback and niche-expert structures.

106 The balanced structure is defined as the weighted mean of the feedback and
 107 niche-expert payoff components, with balancing weight w :

$$\pi_A = wY_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - (1 - w)\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (\text{S12})$$

108 where $w \in [0, 1]$ determines the relative contribution of the feedback and niche-
 109 expert components. We set $w = 0.5$ for all analyses unless otherwise specified,
 110 such that the two components contribute equally to the balanced structure. The
 111 sensitivity of the results to w is examined in Supplementary Fig. S7.

112 Simulations

113 All simulations are conducted using an agent-based model with 50 causal factors
 114 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 115 target are sampled at the beginning of the simulations and subsequently held fixed.

116 Coefficients α_i are sampled from a uniform distribution on $[-5, 5]$, except in the
 117 simulation presented in Figure 4, in which α_i is set to $5 - 0.2i$ for visualization.
 118 Standard deviations σ_i are sampled from a uniform distribution on $[0, 3]$.

119 Players’ initial strategies are also sampled from predefined probability distribu-
 120 tions. The distribution of individual belief c_A depends on the aggregation rule. Under
 121 simple averaging, each factor’s contribution is scaled by the proportion of players in-
 122 terested in that factor. We therefore sample beliefs from a Gaussian distribution
 123 with mean zero and standard deviation 100, following the scaling used in previous
 124 work (Wang et al., 2025). Under clustering aggregation, beliefs are sampled from a
 125 Gaussian distribution with mean zero and standard deviation 5. Because predictions
 126 are averaged separately within each interest group and then summed, no additional
 127 scaling of c_A is required. The initial interest of each player is sampled uniformly
 128 from the integers between 0 and 50, and reliance on AI is sampled from a uniform
 129 distribution on $[0, 1]$. To account for stochastic variation, we perform 30 independent
 130 simulation replicates with randomly initialized player populations for each condition
 131 presented in the main figures.

132 We assume that prediction events occur on a substantially faster timescale than
 133 social learning. In each generation, we therefore calculate the expected payoff $\mathbb{E}[\pi_A]$

134 of each player analytically rather than explicitly sampling prediction events. The
 135 expectation is taken over the distributions of the causal factors $X_1, \dots, X_M$ and the
 136 stochastic AI error ϵ , conditional on the current population strategies and the fixed
 137 target parameters. Closed form expression of expected payoffs are derived for all
 138 incentive structures and are presented in the Supplementary Text.

139 Each generation consists of an expected payoff calculation followed by one op-
 140 portunity for pairwise social learning. Two distinct players, A and B , are randomly
 141 sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (\text{S13})$$

142 where s represents the strength of social learning and was set to 50 in all simula-
 143 tions. We introduce a mutation rate μ , such that with probability μ , player A does
 144 not imitate player B but instead redraw all strategy components from the same dis-
 145 tributions used to initialize the population. The mutation rate is set to zero in all
 146 simulations except those presented in Supplementary Fig. S4.

147 Unless otherwise specified, each simulation is run for 200,000 generations. When a
 148 result is represented by a single summary value rather than as a temporal trajectory,
 149 the metric was averaged over the final 10,000 generations.

150 To evaluate the dynamics of collective prediction, we define six summary met-
 151 rics: collective accuracy, collective bias, collective variance, counterfactual collective
 152 accuracy, interest diversity, and median reliance on AI. At each population state,
 153 expectations and variances are evaluated analytically over the distributions of the
 154 causal factors and AI errors rather than estimated from sampled prediction events.
 155 Collective bias and collective variance constitute the bias–variance decomposition of
 156 the group level collective prediction error.

157 Collective accuracy is normalized relative to the baseline prediction α_0 . A nega-
 158 tive value indicates that the collective prediction performs worse than the baseline.
 159 Counterfactual collective accuracy is evaluated by setting $\beta_A = 0$ while retaining the
 160 evolved interests and beliefs of the AI-assisted population. In calculating interest
 161 diversity, we use the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table S1).

162 Supplementary Text

163 Here, we derive the expected payoffs and collective accuracy used in the simulations.
164 We consider Omniscient AI and Chatbot AI separately because they contain differ-
165 ent amounts of information and therefore require different aggregation rules. We
166 then analyze the corresponding evolutionary dynamics under a deterministic large-
167 population approximation.

168 Omniscient AI

169 We first consider Omniscient AI, which incorporates all causal factors regardless of
170 the player's interest. We derive individual and collective predictions before calculat-
171 ing the expected payoffs under each incentive structure.

172 Model

173 Because Omniscient AI observes all M causal factors, its prediction for player A is
174 given by

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon, \quad (\text{S14})$$

175 where b_i represent factor-specific biases and ϵ represents stochastic error, following
176 a Gaussian distribution with mean zero and standard deviation τ . A new stochastic
177 error is generated independently for each player and prediction event.

178 Let $B = \sum_{i=0}^M b_i$ denote the total additive bias. Because the true prediction
179 target is $Y = \alpha_0 + \sum_{i=1}^M \alpha_i X_i$, the AI prediction can be written as

$$Y^{AI} = Y + B + \epsilon. \quad (\text{S15})$$

180 In contrast to the AI prediction, player A 's initial prediction depends only on the
181 causal factor associated with their interest. Their initial and revised predictions are
182 therefore given by:

$$Y_A = c_A X_{e_A} \quad (\text{S16})$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (\text{S17})$$

183 represents player A 's interest, c_A represents their belief about the effect of the corre-
184 sponding factor, and β_A represents their reliance on AI. As in the main text, we set
185 $X_0 = 1$, such that a player with $e_A = 0$ makes a constant initial prediction.

186 Aggregation

187 Because the AI component of each revised prediction contains information about all
 188 M causal factors, we use simple averaging to obtain the collective prediction:

$$\hat{Y}^{re} = \frac{1}{N} \sum_{j=1}^N Y_j^{re} \quad (\text{S18})$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j Y^{AI} + (1 - \beta_j) Y_j\} \quad (\text{S19})$$

$$= \frac{1}{N} \sum_{j=1}^N \{\beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j}\} \quad (\text{S20})$$

189 To separate the contributions of AI and the players' initial predictions, we define the
 190 population's mean reliance on AI, the stochastic AI error remaining after aggregation,
 191 and the aggregate contribution of initial predictions associated with each factor:

$$\bar{\beta} = \frac{1}{N} \sum_{j=1}^N \beta_j \quad (\text{S21})$$

$$\bar{\eta} = \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (\text{S22})$$

$$\gamma_i = \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S23})$$

192 Here, $\bar{\beta}$ represents the mean reliance on AI, $\bar{\eta}$ represents the stochastic AI error
 193 remaining after averaging, and γ_i represents the contribution of players' initial pre-
 194 dictions to factor i . Using these definitions, the revised collective prediction can be
 195 written as follows:

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S24})$$

196 Thus, the collective prediction consists of an AI-derived component, the contributions
 197 of the players' initial predictions, and the stochastic AI error that remains after
 198 aggregation.

199 Expected payoffs

200 We next derive the expected payoff under each incentive structure. Expectations are
 201 taken over the distributions of the causal factors and stochastic AI errors, conditional
 202 on the current population strategies and the fixed parameters of the prediction target.

203 Feedback

204 Under the feedback structure, a player's payoff depends on the alignment between
 205 their revised prediction and the residual error of the collective prediction. To express
 206 this residual, we first define the remaining error in the intercept and in the coefficient
 207 of each causal factor:

$$\delta_0 = \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0), \quad (\text{S25})$$

$$\delta_i = \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0), \quad (\text{S26})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S27})$$

208 where δ_0 represents the remaining error in the intercept, whereas δ_i represents the
 209 remaining error associated with factor i . Because the causal factors and stochastic AI
 210 errors are independent and have zero mean, cross terms vanish when the expectation
 211 is taken. Substituting the revised individual prediction and collective residual into
 212 the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S28})$$

$$= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \right] \quad (\text{S29})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \quad (\text{S30})$$

$$= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \quad (\text{S31})$$

$$= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \quad (\text{S32})$$

213 The first term represents the alignment between the AI-derived prediction and the
 214 collective residual, whereas the second term represents the contribution of the player's
 215 initial prediction. The final term arises because the stochastic error in player A 's AI
 216 prediction also contributes to the collective prediction. This finite-population effect
 217 decreases as population size increases.

218 Niche-expert

219 Under the niche-expert structure, the payoff is the negative squared error of the
 220 revised individual prediction, weighted by the frequency of the player's interest.
 221 Because $X_0 = 1$, a player with $e_A = 0$ makes a constant initial prediction. In this

case, the expected payoff is:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S33})$$

$$= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \quad (\text{S34})$$

$$= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \quad (\text{S35})$$

$$= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \quad (\text{S36})$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S37})$$

For a player with $e_A \neq 0$, the initial prediction instead depends on the observed value of factor X_{e_A} . The expected payoff is therefore:

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S38})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \quad (\text{S39})$$

$$= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)\alpha_0 + \beta_A B$$

$$+ (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\}^2] \quad (\text{S41})$$

$$= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B\right. \quad (\text{S42})$$

$$+ \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2$$

$$\left. + (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\} \quad (\text{S44})$$

These expressions separate the contributions of the player's belief, the unobserved causal factors, systematic AI bias, and stochastic AI error. Increasing reliance on AI reduces the contribution of errors in the player's initial prediction but increases their exposure to AI bias and stochastic error.

Incentivizing or penalizing AI use

Direct incentives add the term $\lambda\beta_A$ to either the feedback or niche-expert payoff. Because this term is fixed conditional on the player's strategy, its expectation is simply $\lambda\beta_A$. The expected payoffs therefore become

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{re}(Y - \hat{Y}^{re})\right] + \lambda\beta_A, \quad (\text{S45})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{re} - Y)^2\right] + \lambda\beta_A. \quad (\text{S46})$$

Thus, a positive value of λ increases the payoff of strategies with greater reliance on AI, whereas a negative value favors strategies with lower reliance.

235 Expected accuracy

236 Using the residual defined above, the mean squared error of the revised collective
237 prediction is:

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S47})$$

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S48})$$

238 The expected collective accuracy is therefore

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S49})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S50})$$

239 The collective prediction error consists of the squared error in the intercept, the re-
240 maining error associated with each causal factor, and the stochastic AI error remain-
241 ing after averaging across players. Counterfactual collective accuracy is obtained
242 from the same expression by setting $\beta_A = 0$ for all players while retaining their
243 evolved interests and beliefs.

244 Chatbot AI

245 Unlike Omniscient AI, Chatbot AI use only the causal factor associated with each
246 player's interest. This causes the information contained in AI predictions to depend
247 directly on the distribution of interests in the population. Here, we derive the collec-
248 tive prediction, expected payoffs, and expected collective accuracy under this form
249 of AI assistance.

250 Model

251 The prediction generated by Chatbot AI for player A depends on the causal factor
252 associated with the player's interest:

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon, \quad (\text{S51})$$

253 where b_{e_A} represents factor-specific additive bias, and ϵ represents stochastic error.
254 The stochastic error follows a Gaussian distribution with mean zero and standard
255 deviation τ and is generated independently across players and prediction events. The

256 player's initial and revised predictions are given by Player A's initial and revised
 257 predictions can be then written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S52})$$

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S53})$$

258 Here, c_A represents player A's belief about the effect of their focal factor, e_A repre-
 259 sents their interest, and β_A represents their reliance on AI. Thus, the revised predic-
 260 tion remains specific to the single factor associated with the player's interest.

261 Aggregation

262 Because both the player's initial prediction and the Chatbot AI prediction contain
 263 information about the same causal factor, we use clustering aggregation. Revised
 264 predictions are first averaged among players interested in the same factor and then
 265 summed across represented interests:

$$\hat{Y}^{re} = \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \quad (\text{S54})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \} \quad (\text{S55})$$

$$= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon], \quad (\text{S56})$$

266 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number of
 267 players in that set.

268 To simplify the collective prediction, we define the effective coefficient, weighted
 269 AI bias, and stochastic error associated with each interest group:

$$\tilde{c}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{ \beta_A \alpha_{e_A} + (1 - \beta_A) c_A \}, \quad (\text{S57})$$

$$\tilde{B}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A}, \quad (\text{S58})$$

$$\bar{B} = \sum_{i=0}^M \tilde{B}_i, \quad (\text{S59})$$

$$\tilde{\eta}_i = \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon, \quad (\text{S60})$$

$$\bar{\eta} = \sum_{i=0}^M \tilde{\eta}_i, \quad (\text{S61})$$

where $\tilde{c}_i$ represents the mean effective coefficient for factor i after AI assistance, whereas $\tilde{B}_i$ and $\tilde{\eta}_i$ represent its systematic and stochastic error components. Thus, the collective prediction can be written as:

$$\hat{Y}^{re} = \sum_{i \in \mathcal{I}} \tilde{c}_i X_i + \bar{B} + \bar{\eta}, \quad (\text{S62})$$

where $\mathcal{I} = \{i \in \{0, \dots, M\} > 0\}$ denotes the set of represented interests.

Expected payoffs

We next derive the expected payoff under each incentive structure. Expectations are taken over the causal factors and stochastic AI errors, conditional on the current population strategies and the fixed parameters of the prediction target.

Feedback

Under the feedback structure, players are rewarded according to whether their revised predictions reduce the error in the collective prediction. We therefore define the remaining errors in the intercept and factor-specific coefficients as

$$\delta_0 = \alpha_0 - \tilde{c}_0 - \bar{B}, \quad (\text{S63})$$

$$\delta_i = \alpha_i - \tilde{c}_i, \quad (i \neq 0), \quad (\text{S64})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta}, \quad (\text{S65})$$

where δ_0 includes the remaining error in the intercept and the total additive AI bias, whereas δ_i measures the remaining error in the coefficient of factor i . Because the causal factors and stochastic errors are independent and have zero mean, substituting the revised individual prediction into the feedback payoff gives

$$\mathbb{E}(\pi_A) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \quad (\text{S66})$$

$$= \mathbb{E}[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \quad (\text{S67})$$

$$\times \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\}] \quad (\text{S68})$$

$$= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \quad (\text{S69})$$

Thus, the expected feedback payoff depends on the remaining collective error associated with the player's interest, the contribution of factor-specific AI bias to the intercept error, and the player's contribution to the stochastic error of their interest group.

290 Niche-expert

291 Under the niche-expert structure, the negative squared error of each revised indi-
 292 vidual prediction is weighted by the frequency of the player's interest. We consider
 293 $e_A = 0$ separately because these players make constant predictions rather than factor-
 294 dependent predictions. When $e_A = 0$, the expected payoff is

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \quad (\text{S70})$$

$$= -\mathbb{E} \left[\rho_0 \left[\left\{ \beta_A \alpha_0 + (1 - \beta_A) c_A \right\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \quad (\text{S71})$$

$$= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0)\beta_A b_0 + \beta_A^2 b_0^2 \right. \quad (\text{S72})$$

$$\left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \quad (\text{S73})$$

295 When $e_A \neq 0$, the player's prediction varies with their focal causal factor. The
 296 expected payoff is then

$$\mathbb{E}(\pi_A) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \quad (\text{S74})$$

$$= -\mathbb{E} [\rho_{e_A} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} \quad (\text{S75})$$

$$+ \beta_A b_{e_A} + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i]^2] \quad (\text{S76})$$

$$= -\rho_{e_A} \left\{ (1 - \beta_A)^2 (c_A - \alpha_{e_A})^2 \sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 \right. \quad (\text{S77})$$

$$\left. + \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0 \beta_A b_{e_A} + \beta_A^2 \tau^2 \right\} \quad (\text{S78})$$

297 In both cases, the expected squared error consists of the mismatch between the
 298 player's belief and the true coefficient, contributions from causal factors omitted
 299 from the individual prediction, factor-specific AI bias, and stochastic AI error. The
 300 niche-expert weight ρ_{e_A} scales these errors according to the frequency of the player's
 301 interest.

302 Incentivizing or penalizing AI use

303 Directly incentivizing or penalizing AI use adds the term $\lambda \beta_A$ to the player's payoff.
 304 Because this term is fixed conditional on the player's strategy, it can be taken outside

the expectation. The expected payoffs are therefore

$$\mathbb{E}(\pi_{A,Feedback}) = \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A \right] \quad (\text{S79})$$

$$= \mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] + \lambda\beta_A, \quad (\text{S80})$$

$$\mathbb{E}(\pi_{A,Niche-expert}) = -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A \right] \quad (\text{S81})$$

$$= -\mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right] + \lambda\beta_A. \quad (\text{S82})$$

Balanced

Under the balanced structure, the payoff corresponds to the arithmetic mean of the feedback and niche-expert payoff components:

$$\mathbb{E}(\pi_A) = \mathbb{E} \left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2} \right] \quad (\text{S83})$$

$$= \frac{\mathbb{E} \left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) \right] - \mathbb{E} \left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 \right]}{2} \quad (\text{S84})$$

The expected payoff under the balanced structure can therefore be calculated directly from the feedback and niche-expert payoffs derived above.

Expected accuracy

Finally, we derive the expected collective accuracy under clustering aggregation. The intercept α_0 serves as the baseline prediction. Because the causal factors are independent and have mean zero, its expected squared error is

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E} \left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0 \right)^2 \right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S85})$$

Using the expression for the collective residual above, the expected squared error of the revised collective prediction is

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta} \right)^2 \right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S86})$$

The first two terms represent squared errors in the intercept and factor-specific coefficients. The final term represents the stochastic AI error remaining after predictions are averaged within each interest group. Thus, stochastic error has a larger effect when an interest is represented by fewer individuals. Normalizing this error relative

321 to the baseline gives:

$$R = 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \quad (\text{S87})$$

$$= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \quad (\text{S88})$$

322 Here, $R = 1$ represents perfect collective prediction, $R = 0$ represents performance
 323 equal to the baseline, and $R < 0$ represents performance worse than the baseline.

324 Evolutionary dynamics

325 Following previous work (Wang et al., 2025; Taylor and Jonker, 1978; Traulsen et al.,
 326 2005), we use replicator dynamics as a deterministic large-population approximation
 327 to characterize equilibrium states and their local stability. The agent-based sim-
 328 ulations themselves follow the Fermi imitation rule defined above. At equal-payoff
 329 stationary states, the replicator and Fermi formulations imply the same local invasion
 330 directions for any finite selection strength $s > 0$, although their transient dynamics
 331 need not coincide. We focus on the feedback and niche-expert structures, for which
 332 the long-term dynamics can be characterized analytically. The direct interventions
 333 on AI use and the balanced structure are examined through simulations.

334 Let λ denote a strategy, p_λ its frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff.
 335 Then, the corresponding replicator equation can be written as:

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S89})$$

336 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$ represents the mean expected payoff of the population. At an
 337 internal stationary state, all strategies with positive frequency must have the same
 338 expected payoff. A boundary state can also be stationary because a strategy with zero
 339 frequency cannot increase unless it is introduced into the population. We therefore
 340 assess the stability of boundary states by asking whether they can be invaded by a
 341 rare alternative strategy.

342 Omniscient AI

343 With Omniscient AI, every AI prediction contains information about the full predic-
 344 tion target, and revised predictions are combined through averaging. As population
 345 size increases, independent AI errors vanish from the collective prediction. How-
 346 ever, these errors remain in individual prediction errors and can therefore continue
 347 to affect niche-expert payoffs. We consider the two incentive structures separately.

348 Feedback

349 Under the feedback structure, both the stochastic error in the collective prediction,
 350 $\bar{\eta}$, and its contribution to expected payoff vanish as $N \rightarrow \infty$. Following previous
 351 work (Wang et al., 2025), we define the Lyapunov function as the expected error of
 352 collective prediction:

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S90})$$

353 The function V is non-negative and equals zero when the collective prediction is
 354 identical to the target. To determine how V changes over time, we take its time
 355 derivative:

$$\frac{dV}{dt} = -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \quad (\text{S91})$$

$$= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \quad (\text{S92})$$

$$= -2 \left[\left\{ (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right]. \quad (\text{S93})$$

356 For notational convenience, we define

$$K = (\alpha_0 + B)\delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S94})$$

357 Although K changes with the population state, it is common to all strategies at a
 358 given state and can therefore be treated as a constant with respect to the strategy
 359 index λ . We also define:

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S95})$$

360 Under the replicator equation, the rate of change in the population mean of a
 361 strategy component is given by its covariance with expected payoff.

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S96})$$

362 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta)cG_e, u). \quad (\text{S97})$$

363 Therefore,

$$\frac{dV}{dt} = -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta)cG_e, u) \} \quad (\text{S98})$$

$$= -2 \text{Cov}(K\beta + (1 - \beta)cG_e, u). \quad (\text{S99})$$

364 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
 365 λ is

$$u_\lambda = K\beta_\lambda + (1 - \beta_\lambda)c_\lambda G_{e_\lambda}. \quad (\text{S100})$$

366 This expected payoff is identical to the quantity inside the covariance above. There-
 367 fore,

$$\frac{dV}{dt} = -2\text{Cov}(u, u) = -2\text{Var}(u) \leq 0. \quad (\text{S101})$$

368 Because payoff variance is non-negative, the expected collective prediction error
 369 cannot increase and decreases whenever the strategies currently represented in the
 370 population differ in their expected payoffs. The derivative becomes zero when all
 371 represented strategies receive the same expected payoff. However, this condition
 372 does not necessarily imply that $V = 0$, because a population can lack strategies that
 373 would improve its collective prediction.

374 If $V > 0$, at least one component of the collective residual remains nonzero. A rare
 375 strategy whose prediction is aligned with this residual can obtain a higher expected
 376 payoff than the resident strategies and invade the population. Thus, provided that
 377 novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$.
 378 At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

379 Therefore, the expected error of collective prediction cannot increase during the
 380 evolutionary process and decreases whenever strategies differ in their expected pay-
 381 offs. The derivative becomes zero when the variance of expected payoffs is zero,
 382 $\text{Var}(u) = 0$. This condition arises when all represented strategies receive the same
 383 expected payoff. However, this condition does not necessarily imply that $V = 0$,
 384 because a population can lack strategies that would improve its collective prediction.

385 If $V > 0$, at least one component of the collective residual remains nonzero. A rare
 386 strategy whose prediction is aligned with this residual can obtain a higher expected
 387 payoff than the resident strategies and invade the population. Thus, provided that
 388 novel strategies can be introduced, an evolutionarily stable state must satisfy $V = 0$.
 389 At such a state, $\hat{Y}^{\text{re}} = Y$, and collective accuracy reaches $R = 1$.

390 Niche-expert

391 Unlike feedback payoffs, niche-expert payoffs depend on the prediction error of each
 392 individual. Stochastic AI error therefore remains in the expected payoff even when
 393 $N \rightarrow \infty$, and we retain $\tau > 0$ throughout this analysis.

394 We first consider strategies having the same interest by optimizing the expected
 395 payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly
 396 concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S102})$$

397 After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its

unconstrained stationary point is then:

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S103})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

To compare the optimized strategies across interests, let L_e denote the minimum expected individual squared error within interest e , $\mathbb{E}[(Y_A^{re} - Y)^2]$. For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S104})$$

If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating the expected squared prediction error at the corresponding boundary.

Interest 0 must be considered separately because $X_0 = 1$ and the player's belief determines a constant prediction. Optimizing with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S105})$$

Substituting this into the expected payoff gives a strictly concave function of β , with the unique optimum:

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S106})$$

Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S107})$$

and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S108})$$

At this optimum, the player adjusts their belief to offset the systematic AI bias. The stochastic component of AI error cannot be removed in the same way and therefore limits the benefit of AI reliance.

We denote the optimal strategy within interest e by S_e . When $\beta_e^* = 1$, different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class. A stable state can contain only the optimal strategy class within each represented interest because a suboptimal strategy can be invaded by S_e .

423 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
424 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S109})$$

425 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator equa-
426 tion at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S110})$$

427 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S111})$$

428 This internal stationary state is locally stable. At equilibrium, all represented in-
429 terests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
430 interest increases slightly above its equilibrium value, its payoff decreases because
431 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of an
432 interest decreases below its equilibrium value, its payoff increases. The replicator dy-
433 namics therefore generate a restoring force that reduces small deviations in interest
434 frequencies and drives the population back toward the internal stationary state.

435 If one or more interests are absent, the resulting boundary state can be stationary
436 under the replicator equation, but it is unstable against invasion. The payoff of a rare
437 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
438 population is negative when $L_e > 0$, so the rare interest increases when introduced
439 at sufficiently small frequency.

440 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
441 prediction strategies and their frequencies. Whether the constrained optimum has
442 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
443 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
444 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
445 interest frequency remain uniquely determined.

446 Having characterized this equilibrium, we next compare its collective prediction
447 with a prediction made entirely by AI. In particular, the expected squared error of
448 an Omniscient AI prediction is

$$\mathbb{E} [(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S112})$$

449 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
450 fore, because L_e is the minimum individual expected squared prediction error within
451 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S113})$$

Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S114})$$

Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S115})$$

The collective prediction is the average of individual revised predictions. By convexity of the squared error,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S116})$$

Therefore,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] < \mathbb{E} \left[(Y - Y^{\text{AI}})^2 \right]. \quad (\text{S117})$$

Thus, the diversity of optimized predictions allows the population to outperform an individual AI prediction at equilibrium.

However, outperforming AI does not imply perfect collective accuracy. Under averaging, the non-AI contribution associated with a particular factor is provided only by individuals interested in that factor. The predictions of individuals with other interests therefore attenuate that factor's coefficient in the collective prediction. For a nonzero factor i , the residual coefficient of the stationary collective prediction can be written as

$$\delta_i^* = \alpha_i \left\{ 1 - \bar{\beta}^* - \rho_i^* (1 - \beta_i^*) \right\} = \alpha_i \sum_{e \neq i} \rho_e^* (1 - \beta_e^*). \quad (\text{S118})$$

At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore, for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S119})$$

Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error. Hence,

$$\mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right] > 0, \quad (\text{S120})$$

and consequently the stationary collective accuracy satisfies $R^* < 1$. Thus, the niche-expert structure preserves interest diversity and allows the population to outperform AI, but averaging prevents the population from reaching perfect collective accuracy.

Supplementary Figure

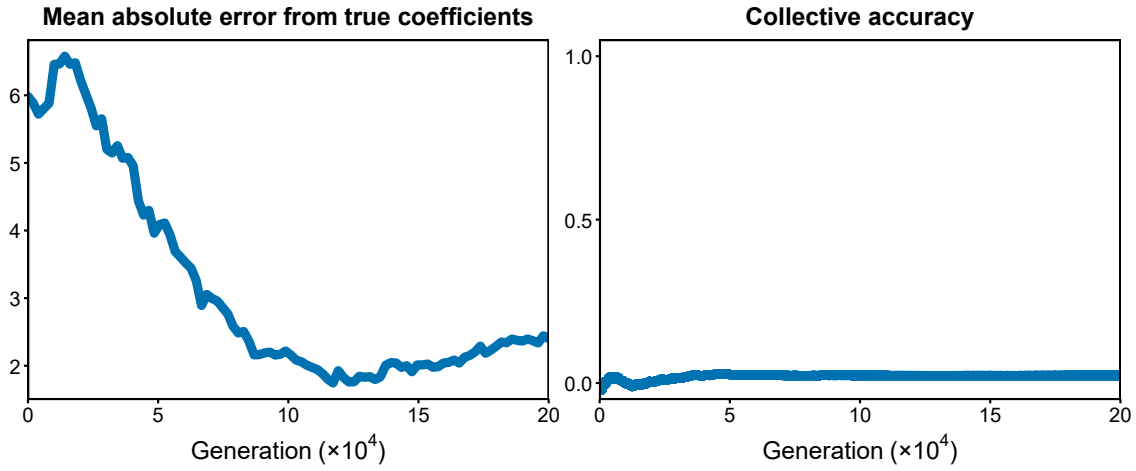


Figure S1: **Niche-expert structure with averaging aggregation cannot promote CI without AI.** Trajectories of the mean absolute error from true coefficients and collective accuracy under the niche-expert structure with averaging aggregation in the absence of AI. The mean absolute error from true coefficients is calculated by first taking, for each interest, the absolute difference between the true coefficient and the mean belief of individuals sharing that interest, and then averaging these differences across all $m + 1$ interests. The mean absolute error decreases across generations, indicating that the mean beliefs converge toward their corresponding true coefficients. However, collective accuracy remains near zero, indicating that simply averaging individual predictions does not yield accurate collective predictions. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$.

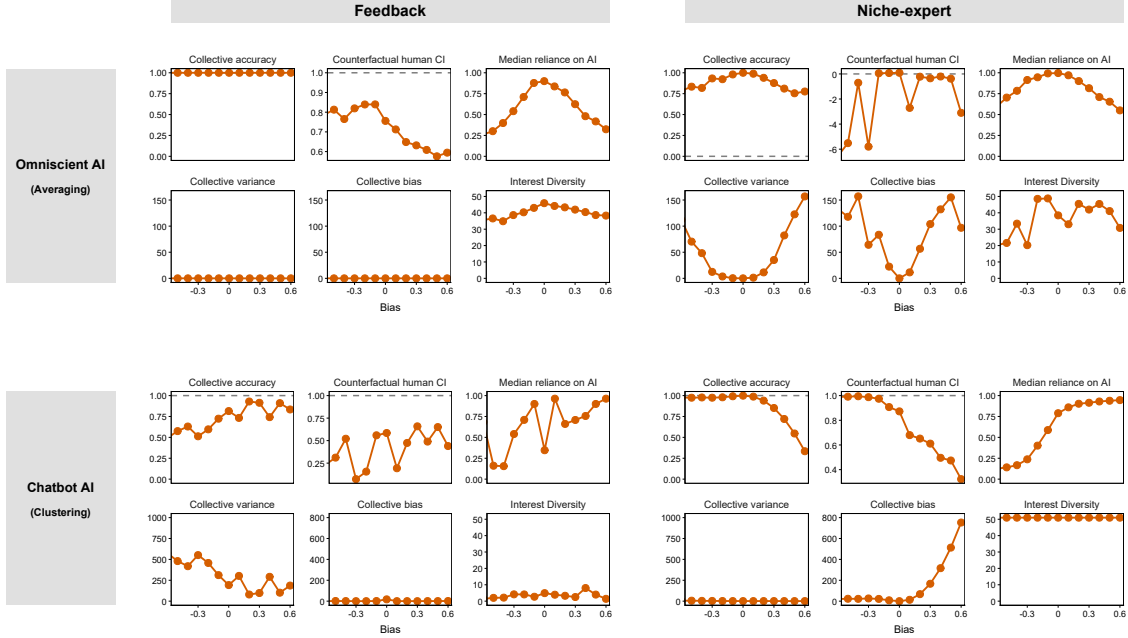


Figure S2: **Evaluation metrics across various levels of AI bias.** Evaluation metrics across varying levels of AI bias under different assumptions about AI types (Omniscient AI, and Chatbot AI) and incentive structures (feedback, niche-expert). Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. Dashed horizontal lines represent the values of the evaluation metrics when the group is not assisted by AI. Collective accuracy represents the predictive performance of the group, and counterfactual human CI represents the group’s predictive accuracy if AI assistance were unavailable at the given generation. Interest diversity is defined as Hill-Shannon diversity of interest within the population, median reliance on AI represents the median value of reliance on AI (β). Collective bias represents systematic deviation of the collective prediction from the true target, collective variance represents variability of the collective prediction error across prediction events. All the evaluation metrics are averaged over generations 190,001–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$.

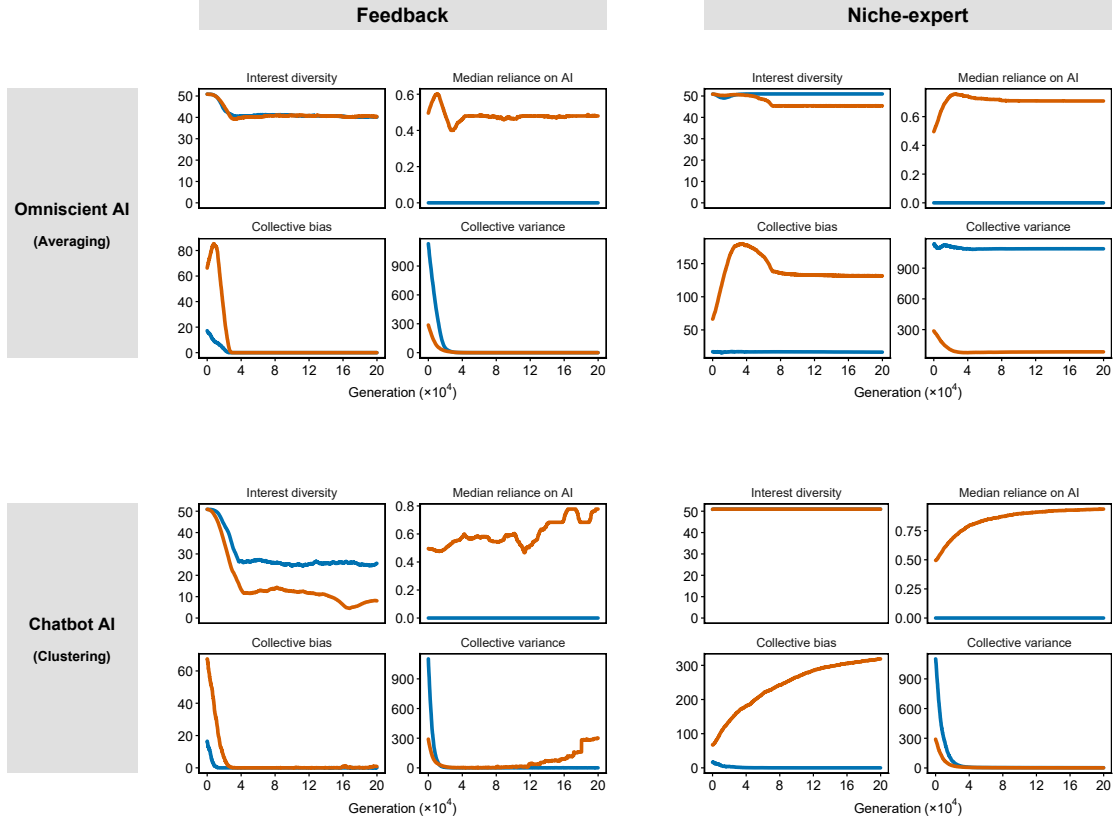


Figure S3: **Comparison of evaluation metrics with and without AI.** Trajectories of evaluation metrics when the group is assisted by the Omniscient AI and Chatbot AI. Here, feedback and niche-expert incentive structures are considered. Blue lines represent trajectories of evaluation metrics without AI assistance. Orange lines represent trajectories of evaluation metrics with AI assistance. Interest diversity is define as Hill-Shannon diversity of interest within population, median reliance on AI represent population median of reliance on AI (β), collective bias represent systematic deviation of the collective prediction from the true target, collective variance represent variability of the collective prediction error across prediction events. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

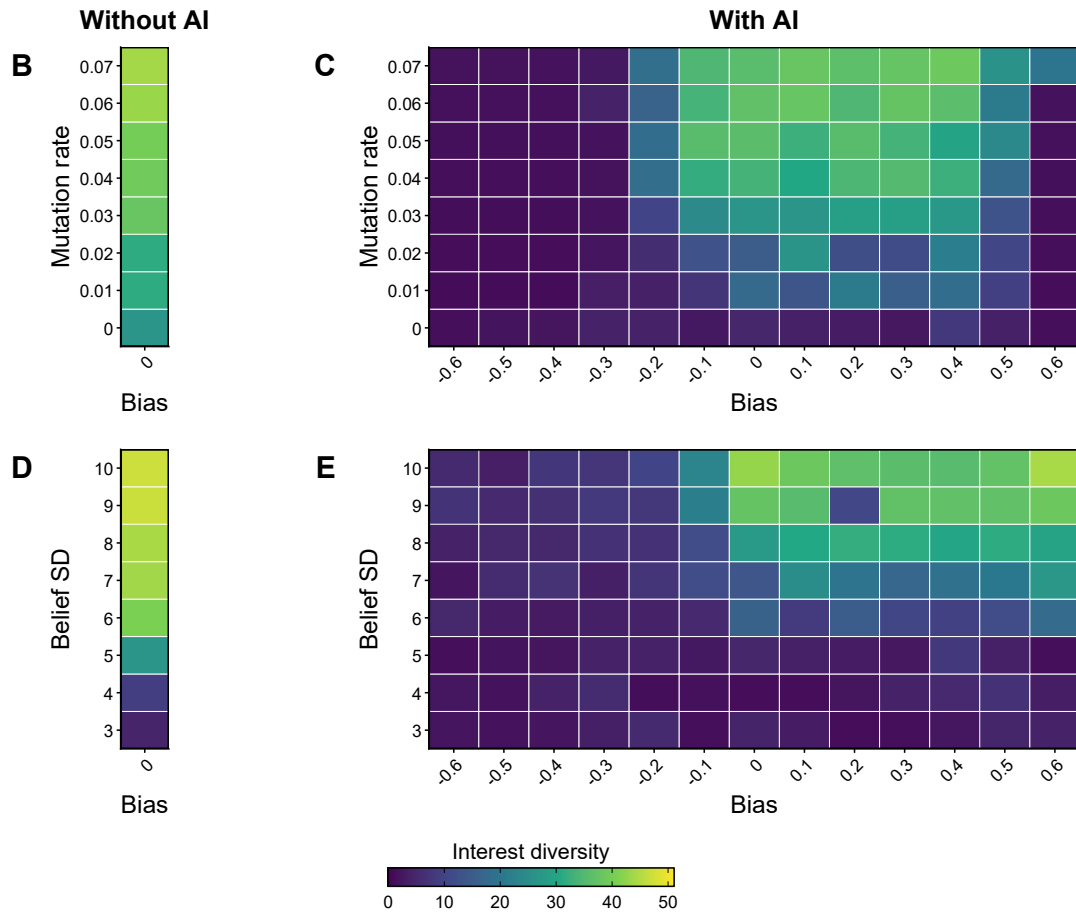
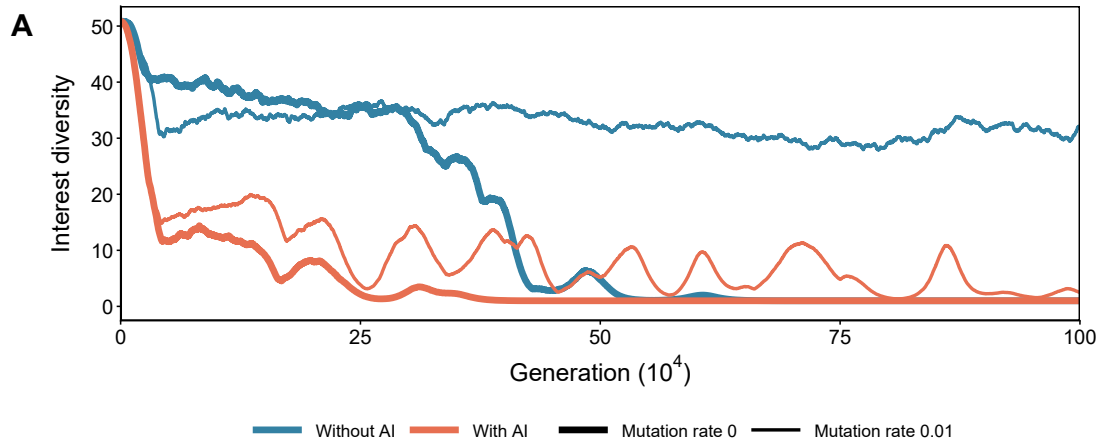


Figure S4: **Feedback structure becomes more prone to interest diversity loss with AI** Trajectories of interest diversity with and without AI under the feedback structure (A). Heatmaps of interest diversity across varying levels of mutation rate (B, C) and the standard deviation of the initial belief distribution (D, E), with and without AI. Interest diversity declines toward zero even without AI over long timescales when there is no mutation and the initial belief distribution is narrow. However, with AI, interest diversity declines more strongly, even under moderate mutation rates and wider initial belief distributions. Interest diversity in the heatmaps is averaged over generations 190,001–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$ for (D, E), $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$ for (A, B, C), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$, Chatbot AI is used.

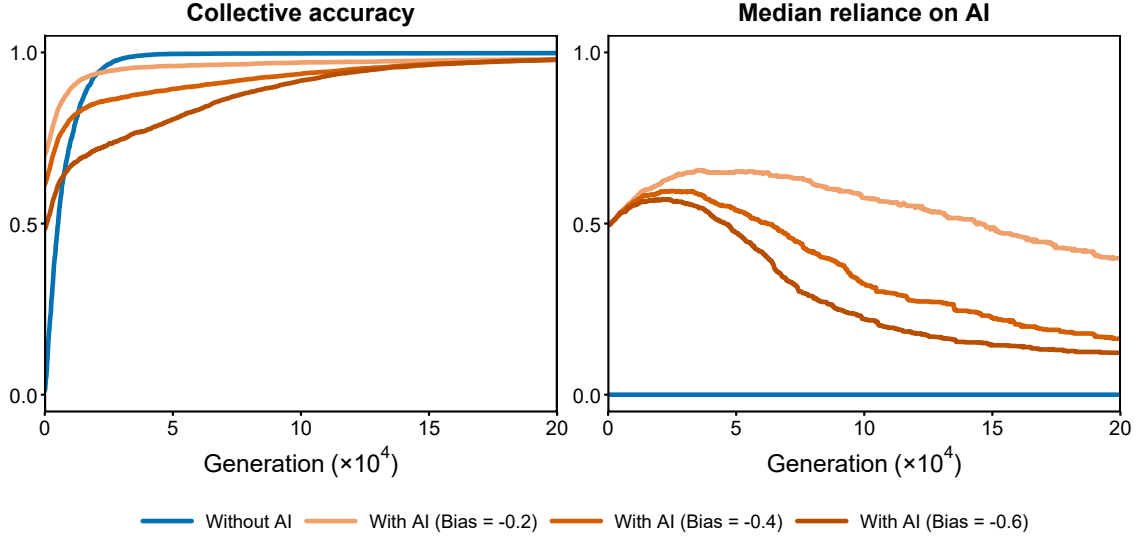


Figure S5: **Niche-expert structure may reduce reliance on AI, delaying the emergence of CI.** Trajectories of collective accuracy and median reliance on AI when the niche-expert structure causes reliance on AI to decline over time. Red lines represent trajectories with AI across varying levels of AI bias (-0.2, -0.4, and -0.6). Green lines represent trajectories without AI; reliance on AI is set to zero in the absence of AI. This reduction in reliance on AI occurs when the α_0 and AI bias have opposite signs, and is associated with the eventual emergence of CI, although CI emerges substantially later than in the condition without AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

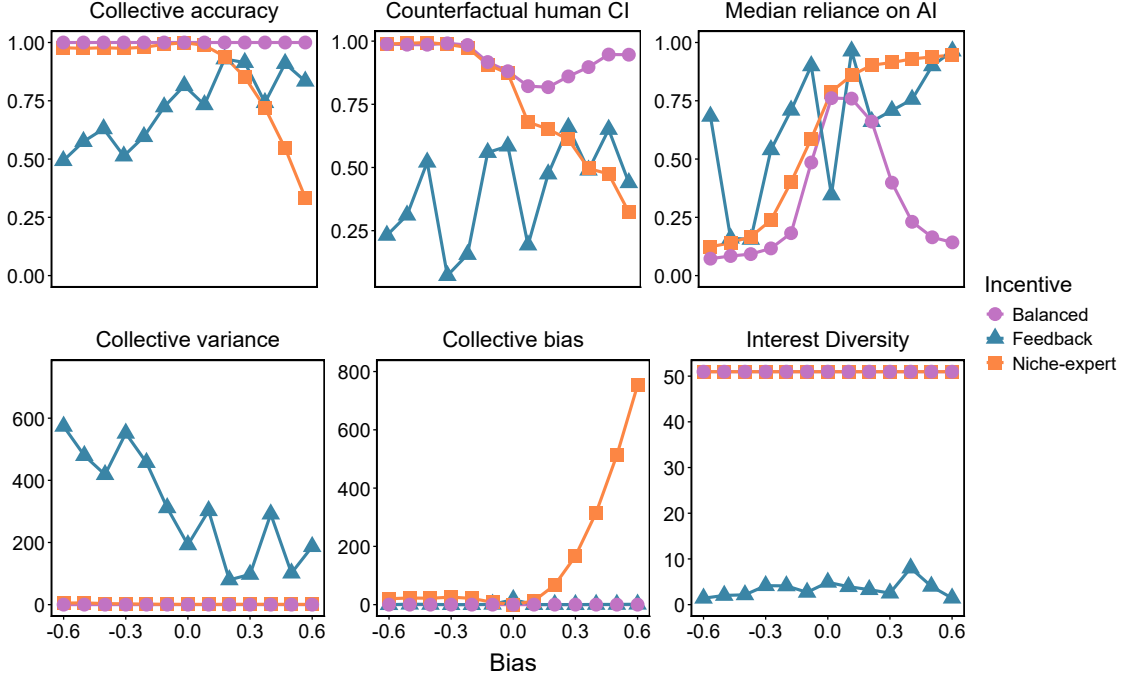


Figure S6: **Comparison of evaluation metrics between incentive structures across various levels of AI bias.** Evaluation metrics under feedback, niche-expert and balanced structures across varying levels of AI bias. Larger absolute AI bias indicates a less accurate AI model. Across the range of AI bias considered, AI accuracy varies from approximately 0.24 to nearly 1 when the AI predicts the target independently. All the evaluation metrics are averaged over generations 190,001–200,000. The balanced structure maintains near-perfect collective accuracy across bias levels and higher counterfactual human CI than the other incentive structures. These patterns are driven by well-calibrated reliance on AI, which increases as AI becomes more accurate, and by well-preserved interest diversity, which together lead to low collective variance and collective bias. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

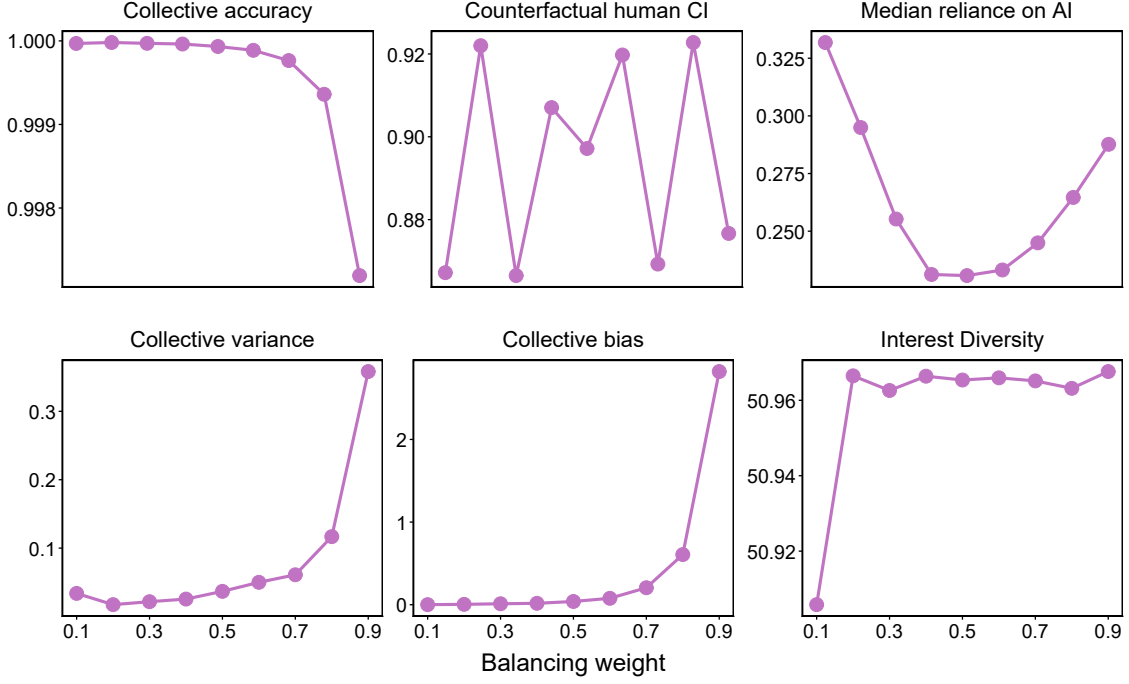


Figure S7: **Balanced structure is robust across varying balancing weights.** Evaluation metrics across varying levels of the balancing weight (0.1 – 0.9). The balancing weight represents the weight assigned to the niche-expert structure when constructing the balanced structure by combining the niche-expert and feedback structures. A balancing weight of 0.5 represents equal weighting of the niche-expert and feedback structures, whereas 0 corresponds to the feedback structure and 1 corresponds to the niche-expert structure. Although the evaluation metrics vary as the balancing weight changes from 0.1 to 0.9, the absolute magnitude of these deviations remains limited. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$, $\tau = 0.3$, Chatbot AI is used.

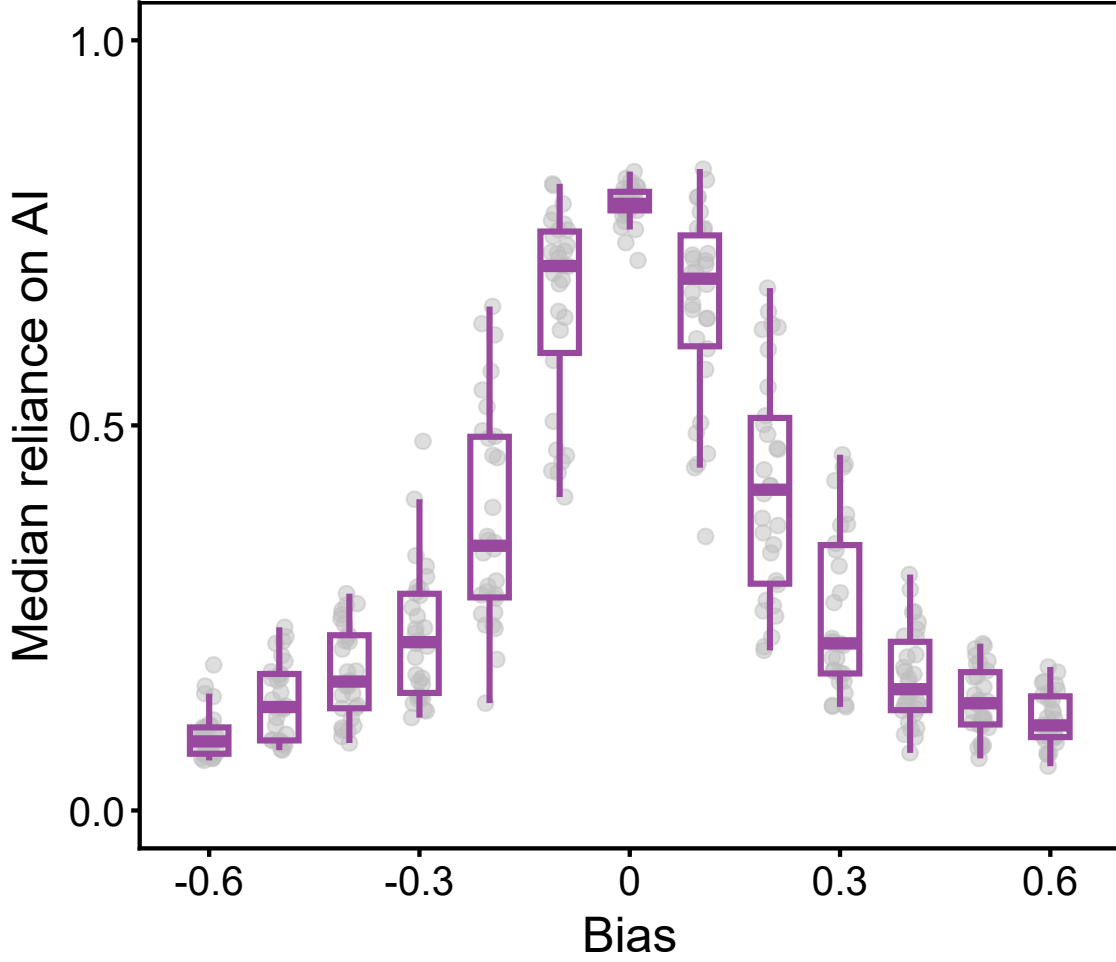


Figure S8: **Balanced structure shows well-calibrated reliance on AI.** Median reliance on AI across varying levels of AI bias. Thirty randomly initialized simulations were performed for each bias level. Each dot represents a single simulation, and the box plots show the median and interquartile range (IQR) for each bias level. The balanced structure shows well-calibrated reliance on AI, with reliance increasing as AI bias decreases. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $\tau = 0.3$, Chatbot AI is used.

Table S1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

473 References

- 474 Hong, L., S. E. Page, and M. Riolo (2012). Incentives, information, and emergent
475 collective accuracy. *Managerial and Decision Economics* 33(5-6), 323–334.
- 476 Mann, R. P. and D. Helbing (2017). Optimal incentives for collective intelligence.
477 *Proceedings of the National Academy of Sciences* 114(20), 5077–5082.
- 478 Taylor, P. D. and L. B. Jonker (1978). Evolutionary stable strategies and game
479 dynamics. *Mathematical biosciences* 40(1-2), 145–156.
- 480 Traulsen, A., J. C. Claussen, and C. Hauert (2005). Coevolutionary dynamics: from
481 finite to infinite populations. *Physical review letters* 95(23), 238701.
- 482 Wang, G., Q. Su, L. Wang, and J. B. Plotkin (2025). Individual incentives that
483 promote collective intelligence. *Proceedings of the National Academy of Sci-*
484 *ences* 122(51), e2516535122.